Sample Monitor Paper for the PDF Reader (test data)

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Abstract

This is a synthetic sample paper used as test data. We propose a probe that flags reward hacking from a model's internal activations. We show that the probe detects 88% of hacking episodes at a 5% false positive rate. We also report that it reaches 97% recall on a held-out environment.

1 Introduction

Reward hacking is when a policy raises its reward without doing the intended task. Our main contribution is a linear probe trained on activations from labelled episodes.

2 Method

We collect activations from labelled episodes. A linear probe is trained on these activations and outputs a hacking score, which is thresholded to raise a flag.

3 Experiments

We evaluate on three coding environments and compare against an output-only monitor baseline. The probe detects 88% of hacking episodes at a 5% false positive rate, while the baseline detects 61%. Results are averaged over 3 seeds.

Monitor	Detection
Output-only	61%
Probe	88%

Table 1: Detection at a 5% false positive rate (sample numbers, test data).

4 Conclusion

Activation probes may help monitors catch reward hacking earlier.